Requirement Analysis:

Project Overview

Hospital Management System is a web-based platform designed to streamline the process of scheduling, managing and tracking medical appointments between patients and healthcare providers. The system replaces manual, phone call based / walk-in appointment booking for convenience. It allows patients to search for doctors, view availability, book or cancel appointments etc.

Problem Statement

Currently, patients seeking medical consultations often face long waiting times, overbooked schedules and inefficient manual appointment processes (phone calls, physical queues or paper registers). Hospital staff struggle to manage doctor availability, avoid double-booking, and handle appointment cancellations or rescheduling in real time. There is no centralized system that allows patients to independently check doctor availability and book appointments, nor one that gives hospital administration a clear overview of scheduling across departments. This leads to patient dissatisfaction, wasted staff time, and inefficient use of medical resources.

Stakeholder Identification

Patients	End users who search for doctors and book/cancel/reschedule appointments
Doctors	Manage their availability, view scheduled appointments, access patient appointment history
Receptionist	Manage appointments on behalf of walk-in/phone patients, oversee daily schedules
Hospital Administrator	Manages doctor accounts, departments, system-wide configuration, and reports

Project scope

In Scope:

- Patient registration and login
- Doctor listing with specialization/department filters
- Appointment booking, rescheduling and cancellation
- Doctor availability/schedule management
- Admin dashboard for managing doctors, departments and appointments
- Appointment status tracking (pending, confirmed, completed, cancelled)

Out of Scope:

- Online payment/billing integration
- Video consultation (telemedicine)

- Insurance claim processing
- Full electronic health record (EHR) management
- Pharmacy/lab integration

Functional Requirements

1. The system shall allow patients to register and create an account with basic personal details.
2. The system shall allow patients to log in and log out securely.
3. The system shall allow patients to search for doctors by name, specialization, or department.
4. The system shall display doctor profiles including specialization, qualifications, and available time slots.
5. The system shall allow patients to book an appointment by selecting a doctor, date and available time slot.
6. The system shall allow patients to view, reschedule or cancel their upcoming appointments.
7. The system shall allow doctors to log in and view their daily/weekly appointment schedule.
8. The system shall allow doctors to update their availability (working hours, leave days).
9. The system shall allow receptionist/admin staff to book, modify, or cancel appointments on behalf of patients.
10. The system shall allow the administrator to add, update, or remove doctor and department records.
11. The system shall prevent double-booking of the same doctor for overlapping time slots

Non-Functional Requirements

1. **Usability**: The interface shall be simple for users with minimal technical expertise.
2. **Performance**: The system shall load pages and return search results within 3 seconds under normal load.
3. **Reliability**: The system shall maintain 99% uptime during business hours.
4. **Security**: Patient data shall be protected using authentication and role-based access control.
5. **Scalability**: The system architecture shall support adding new departments/doctors without major redesign.
6. **Maintainability**: The codebase shall be modular to support future feature additions.
7. **Data Privacy**: Patient personal and medical information shall be handled in compliance with basic data protection principles.

Five use cases

1. **As a patient**, I want to search for doctors by specialization, so that I can find the right doctor for my health concern.

2. **As a patient**, I want to book an appointment with an available doctor, so that I don't have to visit the hospital in person to schedule a visit.
3. **As a doctor**, I want to view and manage my appointment schedule, so that I can plan my day and avoid conflicts.
4. **As a receptionist**, I want to book or modify appointments for walk-in or phone-in patients, so that patients without internet access can still be scheduled.
5. **As a hospital administrator**, I want to manage doctor profiles and department listings, so that the system stays up to date with hospital staffing.

Assumptions and constraints

- Patients and doctors have basic digital literacy and access to a smartphone/computer with internet connectivity.
- The hospital has a fixed, known set of departments and doctors to begin with.
- Appointment durations are treated as fixed-length slots (15 minutes) for simplicity.

Software Process Model Selection

Introduction: The MedCore Hospital Management System is a web-based healthcare management system designed to provide services such as viewing doctors and medical specialties, making appointments, user authentication and providing hospital information. The project consists of several interconnected modules involving both frontend interfaces and backend APIs.

Since the system is developed progressively and requirements may need to be refined during development, selecting an appropriate software process model is important. After studying different software process models, the Spiral Model has been selected for the development of MedCore.

The following software process models were considered:

- Waterfall
- Incremental
- Prototype
- Spiral
- Rapid Application Development (RAD)
- Agile XP
- Scrum
- Kanban

Among these models, the Spiral Model is considered the most suitable for the MedCore project because it combines iterative development with continuous risk identification, evaluation, and refinement.